

Bi-weekly Report

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Group No: 14

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Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

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Objectives for this report

1. Distillation column design and sequencing.
2. Gas-turbine based captive power plant.
3. Cooling water loop.
4. Study-level utility costing, as instructed orally in the first week.

1 Basis

The flowsheet delivered in Report 2 was converged and its balances closed, but the product was not on specification and a few defects were noted at that time. Those items are listed below. The present report takes up all of them, and whatever is still open is given in Section 7.

Table 1: Position at the close of Report 2

Item	Report 2 status	Committed in
Aniline purity, 99.9 wt% required	97.04 wt%	Report 2, Section 5
Water in product, below 0.1 wt%	1.83 wt%	Report 2, Section 5
Nitrobenzene in product, below 5 ppm	660 ppm	Report 1, Section 2
Aniline in wastewater, below 1 ppm	2135 ppm	Report 1, Section 2
Product at 40 °C, 1.0 bar	86.2 °C, 0.2 bar	Report 2, Section 5
COL-1 total condenser on a feed carrying non-condensables	Condenser at −263.15 °C	Report 2, Section 5
ANRECOV returning water to COL-1	101.7 kmol/h returned	Report 2, Section 5
Aniline returned to the reactor in NBBOT	14.59 kmol/h	Report 2, Section 5
MIX-1 pressure specification of zero	Pump absent	Report 2, Section 7
ANILINE–NITRO-01 regression	Not performed	Report 2, Section 6

Two Aspen Plus V14 files have been used. The process flowsheet `che453_R3.12.apw` retains NRTL-RK with HYDRO-01 and NITRO-02 as Henry components under list HC-1 and the ANILINE–WATER pair on the regression case R-DR-1 of Report 1. The captive power plant is a separate file using Peng–Robinson, this being the correct choice for a high-temperature gas-phase combustion system in which no liquid phase and no polar liquid interaction arises. The two are not coupled, the purge being taken as a fixed feed to the power plant.

1.1 Key results

Aniline recovery through the separation train is 98.09 per cent, against 96.5 per cent in Report 2. The product leaves at 40 °C and 1.0 bar with 99.39 wt per cent aniline and 0.04 wt per cent water, against 97.04 and 1.83 wt per cent in Report 2. The selectivities have come back to the design values of Report 1. Total heating is 20.40 MW and total cooling 41.89 MW. The cooling water loop circulates 1578 t/h with 43.6 t/h of make-up. The captive gas turbine gives 709.1 kW net against a plant demand of 708.5 kW. The thermal utility cost works out to USD 9.22 million per year.

2 Distillation Column Design and Sequencing

2.1 The sequence and the reason for it

The sequence is fixed by the aniline–water heterogeneous azeotrope at $x_{\text{aniline}} = 0.0368$ and 99.34 °C, which we obtained from our own regression in Report 1. No single column can separate this pair, because across the two-liquid region the mixture boils at one temperature and gives a vapour of one composition. A decanter breaks the azeotrope by using the miscibility gap directly, so the decanter has to come before the distillation train. This is not a choice, it follows from the phase behaviour.

The remaining choices follow the usual heuristics, which here do not conflict.

- **Free splits first.** The vapour–liquid split is obtained at no separation cost in FLASH and the liquid–liquid split at no separation cost in DEC. Only the residue is handed to distillation.
- **Most plentiful component first.** The organic phase carries 47.9 kmol/h of water against 134.7 kmol/h of aniline, and the water is removed first in COL-1.
- **Hardest split last.** Aniline and nitrobenzene differ by 26.7 K in boiling point and this split carries the 5 ppm nitrobenzene limit. Placed last, its feed is dry and its throughput the smallest of the three, so the column required is the smallest.

The aqueous phase is stripped separately, its treatment being a discharge requirement rather than a product requirement.

2.2 COL-1 — water column

Stages required

COL-1 dries rather than fractionates. At the reboiler the vapour pressure of water is of order 5 bar against 0.13 bar for aniline, and water is strongly positive-deviating in aniline, giving a relative volatility of order 100. Fenske for 99.9 per cent recovery of each key:

$$N_{\min} = \frac{\ln \left[\left(\frac{x_{LD}}{x_{HD}} \right) \left(\frac{x_{HB}}{x_{LB}} \right) \right]}{\ln \alpha} = \frac{\ln [(0.999/0.001)^2]}{\ln 100} = \frac{13.81}{4.605} \approx 3$$

Three theoretical stages at total reflux. The twenty stages of Report 2 were therefore not doing separation work; they were present because the misrouted ANRECOV stream was returning water to the column from which water was being removed, so the same duty was being performed twice. Twelve stages have been adopted, four times the minimum, which leaves margin for stage efficiency.

Condenser

A total condenser must produce a fully liquid distillate. The COL-1 overhead carries hydrogen and nitrogen, which are Henry components precisely because they are supercritical here and possess no condensation temperature. No solution exists and the solver walks to its lower flash bound of 10 K,

which is the -263.15 °C reported. The -12.888 MW of Report 2 is not a condenser duty; it is the latent duty plus the sensible heat of chilling the whole condensate to 10 K.

A Partial-Vapor-Liquid condenser has been used, which produces a vapour and a liquid distillate natively and requires only the ratio between them. The non-condensables leave in VENT. Three values of the distillate vapour fraction were examined.

Table 2: Selection of the distillate vapour fraction on COL-1

Distillate vapour fraction	Condenser T , °C	Condenser duty, MW	Outcome
0.037	55.7	-7.24	Converged; approach to CW only 10.7 K
0.060	71.0	-6.86	Converged
0.100	—	—	Failed to converge on Report 2 feed
0.100	66.9	-2.76	Converged after the reroute; adopted

At 0.10 on the original feed the run terminated with SOLVER01 failing to converge. The cause is that withdrawing too large a proportion of the overhead as vapour strips the top stage of liquid and the top-stage material balance runs short. After the ANRECOV reroute of Section 2.3 the feed is much lighter and 0.10 converges. It has been adopted because it gives the largest approach to the cooling water, which matters in Section 4.

Drying specification and the steam level

Water leaving in the COL-1 bottoms follows the aniline into the COL-2 distillate and appears directly in the product, so the bottoms water content controls the product water specification. A design specification has been set on the water mole fraction in DRYORG at 0.002, manipulating the distillate rate over 50 to 75 kmol/h. It converged at 60.19 kmol/h with $x_{\text{H}_2\text{O}} = 0.00200$.

Design specifications have been preferred to sensitivity analyses throughout, for the reason in Section 9.

The consequence of drying properly is that the reboiler rises from 151.89 °C to 185.95 °C, this being the bubble point of essentially dry aniline at 1.099 bar. Saturated steam at 10 bar is available at 184 °C and cannot drive it. COL-1 has therefore been placed on 16 bar steam at 201 °C. This is unavoidable if the water specification is to be met and it is carried into the costing of Section 6.

Aniline lost overhead: why reflux cannot recover it

A second design specification was attempted on aniline mole recovery to the bottoms at 0.995, manipulating the reflux ratio over 2 to 8. It failed with

DESIGN SPEC IS NOT SATISFIED BECAUSE ONE OR MORE
MANIPULATED VARIABLE IS AT ITS BOUND

The reason is thermodynamic. The aniline appearing overhead is not there through insufficient rectification but because the overhead composition sits at the heterogeneous azeotrope. In the converged case WATOVHD carries $x_{\text{aniline}} = 0.0464$ against an azeotropic composition of 0.0368, so the column is already operating against the azeotrope and no reflux ratio can pass it. The specification was chasing an unreachable target and was deleted.

The column was then returned to Vapor-Liquid-Liquid operation with WATER as the key component for the second liquid phase, so that the two phases actually formed on condensation are represented. Report 2 declared three phases over stages 2–19; here it is restricted to stage 1 alone, since that is where the split occurs, and the restriction reduces the solver load appreciably, which addresses the slow convergence of COL-1 noted in Report 2 Section 7. The effect was to raise the condenser from 55.7 °C

to 66.9 °C and to reduce both duties slightly. It did not materially reduce the aniline lost overhead, which remains governed by the azeotrope; that loss is quantified in Section 3.3 and carried to Section 8 as outstanding.

It was intended to run this case on the Strongly Non-Ideal Liquid algorithm, but Aspen Plus locks the algorithm to Standard whenever design specifications are active on the block. Standard proved adequate with three-phase behaviour confined to one stage.

Table 3: COL-1, Report 2 against Report 3

	Report 2	Report 3
Number of stages	20	12
Feed stages	ORG 10, ANRECOV 2	ORG 4
Condenser	Total	Partial-Vapor-Liquid
Valid phases	VLL, stages 2–19	VLL, stage 1
Distillate vapour fraction	—	0.10
Operating specifications	D 150 kmol/h, RR 3	$x_{\text{H}_2\text{O},B} = 0.002$, RR 3
Top / reboiler pressure, bar	1.0 / 1.1	1.0 / 1.099
Distillate, kmol/h	150.00	60.19
Bottoms, kmol/h	158.42	134.29
Boilup ratio	5.45	2.25
Condenser temperature, °C	−263.15	66.91
Condenser duty, MW	−12.888	−2.763
Reboiler temperature, °C	151.89	185.95
Reboiler duty, MW	+11.076	+3.896
Water in DRYORG, mol %	9.0	0.20
Steam level	10 bar	16 bar

2.3 Rerouting of the recovered aniline

ANRECOV was returned to COL-1 in Report 2 and carried water into the column. It has been routed to DEC instead, where the decanter performs the same phase separation at no energy cost.

ANRECOV leaves the stripper as vapour at 97.5 °C and 1.0 bar whereas DEC operates at 40 °C and 1.5 bar, so a condenser ARCOND has been introduced at 40 °C and 1.5 bar. It is specified liquid-only. A pump placed downstream of a two-phase specification failed with

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NET POSITIVE SUCTION HEAD IS LESS THAN ZERO
FLASH CALCULATIONS FAILED TO CONVERGE IN 30 ITERATIONS
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because the stream carries hydrogen, nitrogen and ammonia and retains a small vapour fraction at 40 °C. Specifying the block liquid-only removes the difficulty and absorbs the small pumping duty; a pump below 0.2 kW belongs at this point in the physical plant and is recorded as an equipment item although it is not a separate block.

The first attempt at the rerouted case also failed inside COL-1 with

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THE FOLLOWING STAGES DRIED UP (VAPOR OR LIQUID FLOW APPROACHES 0):  8 9 11
COMPONENT BALANCE EQUATION FAILED TO CONVERGE
BLOCK COL-1 IS NOT IN MASS BALANCE, RELATIVE DIFFERENCE = 0.229
```

The cause was the operating specification rather than the topology. The column was specified on a fixed bottoms rate of 158 kmol/h while its feed had fallen from 308 to about 205 kmol/h, so during the

recycle iterations the feed passed below the bottoms rate, the distillate went negative and the stages ran dry. Specifying the distillate rate instead, and allowing the bottoms to follow by difference, resolved it.

The rerouting is not free. The COL-1 reboiler falls by 7.18 MW, but a condensing duty of 7.659 MW appears at ARCOND. So the total duty is about the same. What changes is the grade: a duty which earlier needed 16 bar steam now goes to cooling water, and on the prices of Section 6 that swap is worth about \$334 per hour. The other effect is that the circulation around the decanter and the stripper goes up a lot, the stripper feed rising from 244 to 851 kmol/h, which is taken up in Section 2.5.

2.4 COL-2 — aniline column

The feed changed entirely with the COL-1 redesign, the water in DRYORG falling from 9.0 to 0.20 mol per cent, so the column was re-specified rather than re-tuned.

The defect lay in the operating specification. A distillate-to-feed ratio of 0.9 forces the bottoms to be one tenth of the feed irrespective of how much nitrobenzene is present. The feed carries 1.32 kmol/h of nitrobenzene, so a bottoms of 14.6 kmol/h is oversized elevenfold and must carry aniline with it. Replacing the ratio by a bottoms rate of 2 kmol/h reduced the aniline returned to the reactor from 13.39 to 0.831 kmol/h.

This matters beyond the separation train. Both side reactions are specified on aniline, so aniline returned in NBBOT re-enters the reactor and is partly destroyed. Removing it restored the selectivity to the design values, as shown in Section 3.2.

Two further changes were tested and both rejected on the evidence.

Table 4: COL-2 parameters tested and rejected

Change	Gain	Cost	Decision
Reflux ratio 2 $\rightarrow$ 4	NB 713 $\rightarrow$ 295 ppm	Reboiler duty +74 %	Rejected
Bottoms 2 $\rightarrow$ 3 kmol/h	NB 1466 $\rightarrow$ 1230 ppm	Aniline recovery 99.4 $\rightarrow$ 98.6 %	Rejected

In the first case the purity scarcely moved despite the duty rise, because at that point cyclohexylamine and water rather than nitrobenzene were the binding impurities; the reflux was being spent on the wrong separation. In the second the nitrobenzene fell only 16 per cent while a further 0.97 kmol/h of aniline was pushed into the recycle. Both were reverted.

Table 5: COL-2, Report 2 against Report 3

	Report 2	Report 3
Number of stages, feed stage	15, 6 on-stage	15, 6 on-stage
Operating specifications	$D:F$ 0.9, RR 2	Bottoms 2 kmol/h, RR 2
Top / reboiler pressure, bar	0.2 / 0.3	0.2 / 0.3
Distillate, kmol/h	142.58	132.29
Bottoms, kmol/h	15.84	2.00
Aniline in bottoms, kmol/h	14.591	0.831
Bottoms composition	92 mol % aniline	58 mol % nitrobenzene
Condenser temperature, °C	—	129.10
Condenser duty, MW	−6.490	−5.322
Reboiler temperature, °C	144.15	153.15
Reboiler duty, MW	+5.959	+4.852

Why the light-ends draw proposed in Report 2 does not work

Report 2 proposed a light-ends draw as the route to 99.9 wt per cent, both water and cyclohexylamine being lighter than aniline. Since COL-1 now removes the water, cyclohexylamine is the only remaining light end of consequence, and the natural place to take it is COL-1, where an intermediate boiler between water at 100.0 °C and aniline at 184.1 °C should concentrate. Two draws were tested.

Table 6: Cyclohexylamine side draw trials on COL-1

Location	Rate, kmol/h	CHA in draw, mol %	Draw composition
Stage 8	2.0	0.21	67.5 mol % water, 32.0 mol % aniline
Stage 6	1.0	0.12	Water and aniline

Both draws removed water and aniline and left the cyclohexylamine untouched; the stage 8 draw carried 0.64 kmol/h of aniline out of the column for 0.004 kmol/h of cyclohexylamine removed, and cyclohexylamine in DRYORG did not fall. The composition profile shows no intermediate peak anywhere in COL-1; cyclohexylamine tracks the aniline down the column. Both draws were deleted, since they removed product without removing the impurity.

The reason is that boiling point alone is a poor guide here. Cyclohexylamine and aniline are both amines of similar polarity and similar molar mass, so even with a 49.6 K difference in boiling point their relative volatility in the aniline-rich liquid is close to one. It is the same kind of effect as the aniline–water azeotrope: what the vapour pressures suggest is misleading, and the activity coefficients decide the outcome.

Two routes remain. A dedicated light-ends column on DRYORG between COL-1 and COL-2 is a fourth column and an addition to the capital. The better route is at source: cyclohexylamine is formed by over-hydrogenation of aniline at 1.5 per cent selectivity, and a more selective catalyst or a reduced hydrogen partial pressure would suppress it and improve the yield at the same time. This connects directly to the kinetic model due in Report 4.

2.5 STRIP — wastewater stripper

The aqueous phase rose from 244 to 851 kmol/h with the ANRECOV rerouting, the water now leaving through the stripper rather than the COL-1 overhead. At the boilup ratio of unity carried from Report 2 the residual aniline rose to 2.6 wt per cent, and the boilup ratio was studied as a series of discrete converged cases.

Table 7: Effect of the stripper boilup ratio

Boilup ratio	Reboiler duty MW	Overhead kmol/h	Aniline in WASTEWTR kmol/h	Aniline ppm mass
1.0, as Report 2	2.62	187	1.237	26,200
3.0, adopted	7.97	615	0.0073	154
5.0	13.06	1025	0.0027	57

From unity to three the residual falls by a factor of 177. From three to five it falls by a factor of 2.7 only, at the cost of a further 5.09 MW of reboiler duty and a further 410 kmol/h of overhead returning to the decanter and thence to COL-1. Neither case meets the 1 ppm discharge limit and both therefore require the same polishing step, so the case of lower duty has been adopted. On the prices of Section 6 the difference is \$132 per hour, that is \$1.06 million per annum.

The residual of 154 ppm is a reduction by a factor of 13.9 on the 2135 ppm of Report 2 but remains two orders of magnitude above the limit. Equilibrium stripping cannot reach 1 ppm for this system at

any practicable boilup, and a polishing step, either activated carbon adsorption or biological treatment, shall be required. This is a limitation of the unit operation, not of the model.

The wastewater also carries 1.637 kmol/h of cyclohexylamine, that is 3.7 wt per cent, which is a larger discharge problem than the aniline and is recorded in Section 8.

2.6 Hydraulic completion and product conditioning

Table 8: Blocks added

Block	Model	Specification	Result
ARCOND	Heater	40 °C, 1.5 bar, liquid only	−7.659 MW
P-1	Pump	0.2 to 1.0 bar, $\eta = 0.75$, driver 0.95	0.392 kW, head 8.34 m
PRODCOOL	Heater	40 °C, 1.0 bar	−0.664 MW
P-2	Pump	0.3 to 3.0 bar, $\eta = 0.75$, driver 0.95	0.023 kW, head 27.03 m

P-2 closes the defect of Report 2 Section 7, MIX-1 having adopted the 0.3 bar of the COL-2 bottoms with VAP-1 thereafter repressurising to 3 bar without a pump. The MIX-1 pressure specification has been set to 3.0 bar accordingly. P-1 and PRODCOOL bring the product from 129.10 °C and 0.2 bar to 40 °C and 1.0 bar as specified.

One point for detailed design: the pump discharge computes at 129.13 °C against a condenser temperature of 129.10 °C, so the pump takes suction at the bubble point and the available NPSH is nominal. Either a few degrees of subcooling shall be taken before the pump or adequate static head provided at the reflux drum.

3 Material and Energy Balance

3.1 Overall material balance

Table 9: Overall material balance

In	kmol/h	kg/h	Out	kmol/h	kg/h
NB-FEED	141.28	17,393.2	PRODUCT	132.29	12,307.2
H2-MAKE	444.00	906.6	WATOVHD	54.17	1,306.1
			WASTEWTR	235.44	4,374.9
			VENT	6.02	262.6
			PURGE	9.79	49.0
Total		18,299.7	Total		18,299.8

The balance closes to 0.01 kg/h in 18,299.7 kg/h, that is 0.00006 per cent, against 0.7 kg/h in Report 2. The fresh nitrobenzene of 141.28 kmol/h and hydrogen make-up of 444.0 kmol/h are unchanged from Report 2 and agree with the hand balance of Report 1.

3.2 Reactor performance and the hydrogen loop

The selectivities have returned to the design values of Report 1, which the reduction of the aniline recycle in Section 2.4 was intended to achieve:

$$S_{\text{AN}} = \frac{133.911}{141.034} = 94.95 \%, \quad S_{\text{CHA}} = \frac{2.160}{141.034} = 1.53 \%, \quad S_{\text{BZ}} = \frac{4.963}{141.034} = 3.52 \%$$

against the design split of 95 / 1.5 / 3.5 per cent and against 94.47 per cent achieved in Report 2. The

Table 10: Reactor performance against Report 1 and Report 2

Quantity	Report 1, hand	Report 2	Report 3
Nitrobenzene to reactor, kmol/h	142.71	142.54	142.46
Nitrobenzene reacted, kmol/h	141.28	141.11	141.03
Conversion per pass, %	99.0	99.0	99.0
Hydrogen at reactor inlet, kmol/h	1427.1	1300.6	1461.8
Hydrogen-to-nitrobenzene ratio	10.00	9.12	10.26
Hydrogen consumed, kmol/h	435.15	435.88	434.54
Net aniline produced, kmol/h	134.22	133.31	133.91

hydrogen-to-nitrobenzene ratio of 10.26 is also closer to the Report 1 assumption of 10 than the 9.12 obtained in Report 2, the recycle gas now settling at 90.2 mol per cent hydrogen and 4.5 mol per cent nitrogen.

The purge fraction remains as derived in Report 1, the hydrogen leaving in the purge equalling the make-up less what reacts and less what dissolves:

$$f_{\text{purge}} = \frac{M - C - D}{V} = \frac{443.556 - 434.543 - 0.128}{1027.105} = 0.00865$$

against the 0.0086 entered in SPLIT.

3.3 Aniline accounting

Table 11: Disposition of the aniline produced in the reactor, 133.911 kmol/h

Destination	kmol/h	kg/h	% of production
PRODUCT	131.351	12,232.5	98.09
WATOVHD	2.515	234.2	1.88
VENT	0.031	2.8	0.02
WASTEWTR	0.007	0.7	0.01
PURGE	0.008	0.7	0.01

Recovery through the separation train is 98.09 per cent against 96.5 per cent in Report 2. The dominant remaining loss is WATOVHD at 234 kg/h, discussed in Section 8; the 0.831 kmol/h in NBBOT is not a loss, being returned to the reactor.

3.4 Component molar flows, reaction section (kmol/h)

	NB-FEED	H2-MAKE	RXFEED	RXOUT	VAP-OUT	RECYCLE	PURGE	LIQ-OUT
Aniline	—	—	1.709	135.620	0.885	0.877	0.008	134.735
Nitrobenzene	141.280	—	142.458	1.425	0.010	0.010	tr	1.415
Water	—	—	40.053	322.121	40.407	40.059	0.347	281.714
Benzene	—	—	17.211	22.174	17.358	17.208	0.149	4.816
Cyclohexylamine	—	—	0.020	2.180	0.020	0.020	tr	2.160
Nitrogen	—	0.444	50.822	50.822	50.815	50.378	0.437	0.007
Hydrogen	—	443.556	1461.776	1027.233	1027.105	1018.272	8.833	0.128
Ammonia	—	—	2.072	7.035	2.091	2.073	0.018	4.944
Total	141.280	444.000	1716.121	1568.609	1138.690	1128.897	9.793	429.919

3.5 Component molar flows, separation section (kmol/h)

	ORG	AQ	ANRECOV	VENT	WATOVHD	DRYORG	PRODUCT	NBBOT	WASTEWTR
Aniline	134.728	15.308	15.301	0.031	2.515	132.183	131.351	0.831	0.007
Nitrobenzene	1.415	0.024	0.024	0.001	0.099	1.315	0.147	1.169	—
Water	47.919	793.864	560.068	1.537	46.113	0.269	0.269	—	233.795
Benzene	4.816	0.154	0.154	2.586	2.230	tr	tr	—	—
Cyclohexylamine	0.523	2.037	0.401	tr	tr	0.523	0.523	—	1.637
Nitrogen	0.007	0.004	0.004	0.007	—	—	—	—	—
Hydrogen	0.128	0.112	0.112	0.128	—	—	—	—	—
Ammonia	4.944	39.422	39.422	1.729	3.215	—	—	—	—
Total	194.480	850.925	615.486	6.019	54.172	134.289	132.289	2.000	235.439

Two observations from these tables affect equipment sizing. Benzene is produced at 4.963 kmol/h per pass but 17.21 kmol/h circulates in the gas loop, and ammonia is produced at 4.963 kmol/h but 39.42 kmol/h circulates in the aqueous loop. Both close at steady state, benzene leaving as 2.586 in VENT, 2.230 in WATOVHD and 0.149 in PURGE, and ammonia as 1.729, 3.215 and 0.018 respectively, but the circulating inventories are three to eight times the production rates and it is the inventories which size the equipment.

3.6 Stream conditions

3.7 Product stream

The contained aniline is 12,232.5 kg/h, that is 12.23 t/h against a design rate of 12.5 t/h and 97,860 t/y against a design basis of 100,000 t/y, or 97.9 per cent of capacity. Report 2 delivered 11.98 t/h.

The water content of 0.04 wt per cent is within the 0.1 wt per cent limit, having been closed upstream in COL-1 rather than in the product column, and the delivery conditions of 40 °C and 1.0 bar follow from the conditioning of Section 2.6. The residual 0.61 wt per cent of impurity is chiefly cyclohexylamine, for the reason established in Section 2.4. The nitrobenzene content of 1466 ppm is higher than the 660 ppm of Report 2, this being the accepted consequence of reducing the bottoms cut: the trade is 12.6 kmol/h of aniline recovered against 0.08 kmol/h of additional nitrobenzene admitted.

3.8 Energy balance

Taking FEHE as an internal transfer, the block duties and shaft work sum to −21.211 MW against −21.210 MW obtained from the enthalpy flows of the terminal streams, so the energy balance closes to 0.005 per cent. Total heating is 20.40 MW and total cooling 41.89 MW, against 22.48 and 45.16 MW in Report 2.

Three points follow. The stripper reboiler at 7.972 MW is now the largest single heating load, exceeding the COL-1 reboiler twofold; the ANRECOV rerouting moved the burden rather than eliminating it, and moved it from a 16 bar to a 10 bar steam service, which is the gain. The reaction exotherm of 15.354 MW at 300 °C against a heating requirement of 20.40 MW leaves the plant short by approximately 5 MW even under perfect integration. And heat integration remains nominal: FEHE recovers only 0.211 MW because it heats the nitrobenzene vapour alone at 143 kmol/h while 1573 kmol/h of hydrogen joins downstream at MIX-2, so 92 per cent of the reactor feed by mole bypasses the exchanger and RXFEED reaches only 153 °C. This is the subject of Report 5.

4 Cooling Water Loop

4.1 Assignment of services

A service is placed on cooling water where its heat is rejected too low to be recovered usefully. Three are excluded, for two different reasons.

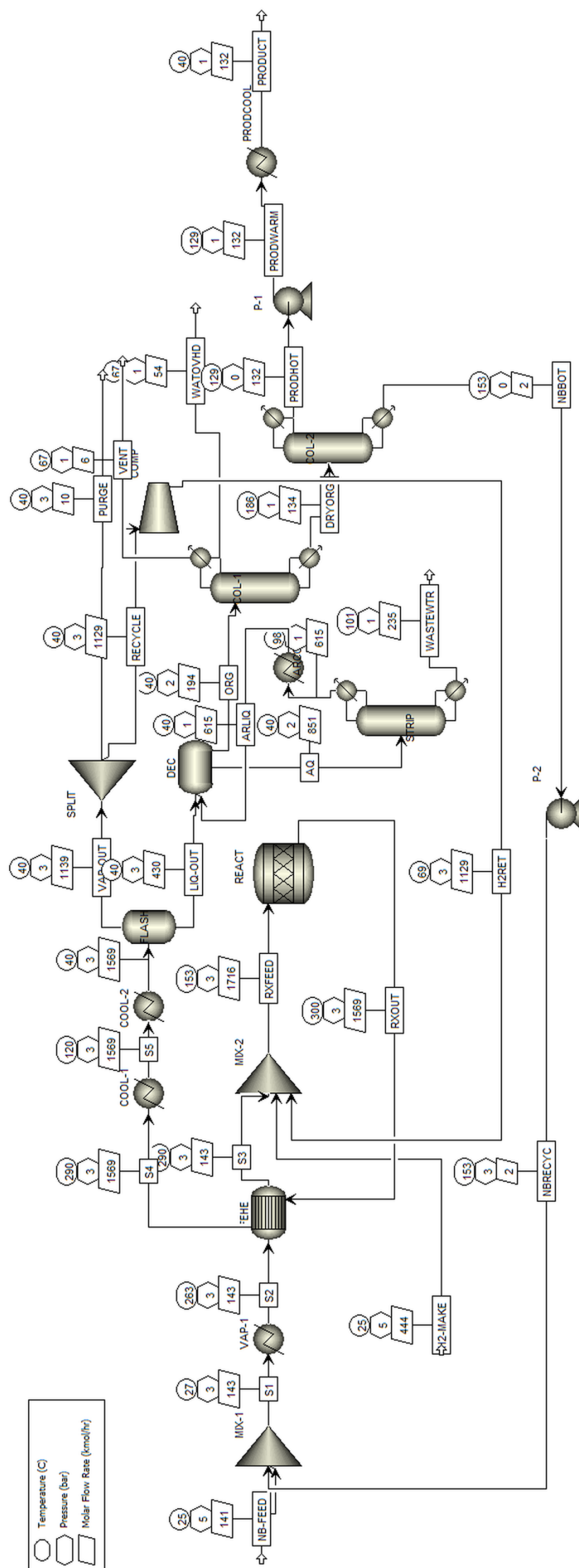


Figure 1: Process flow diagram as converged, file `che453_R3_12.apw`, annotated with temperature (°C), pressure (bar) and molar flow (kmol/h). ARCOND, P-1, P-2, PRODCOOL and the stream VENT are new in this report.

Stream	T , °C	P , bar	Vapour fraction	kmol/h	kg/h
NB-FEED	25.00	5.00	0.000	141.28	17,393.2
NBRECVC	153.25	3.00	0.000	2.00	221.3
S1	27.12	3.00	0.000	143.28	17,614.4
S2	263.05	3.00	1.000	143.28	17,614.4
S3	290.00	3.00	1.000	143.28	17,614.4
H2-MAKE	25.00	5.00	1.000	444.00	906.6
H2RET	69.41	3.20	1.000	1128.84	5,650.1
RXFEED	153.33	3.00	0.987	1716.12	24,171.1
RXOUT	300.00	3.00	1.000	1568.61	24,171.1
S4	289.66	3.00	1.000	1568.61	24,171.1
S5	120.00	2.80	0.945	1568.61	24,171.1
S6	40.00	2.60	0.725	1568.61	24,171.1
VAP-OUT	40.00	2.50	1.000	1138.69	5,699.1
PURGE	40.00	2.50	1.000	9.79	49.0
RECYCLE	40.00	2.50	1.000	1128.90	5,650.1
LIQ-OUT	40.00	2.50	0.000	429.92	18,472.0
ANRECOV	97.50	1.00	1.000	615.49	12,241.2
ARLIQ	40.00	1.50	0.000	615.49	12,241.2
ORG	40.00	1.50	0.000	194.48	14,097.2
AQ	40.00	1.50	0.000	850.92	16,616.1
VENT	66.91	1.00	1.000	6.02	262.6
WATOVHD	66.91	1.00	0.000	54.17	1,306.1
DRYORG	185.95	1.099	0.000	134.29	12,528.5
WASTEWTR	101.15	1.05	0.000	235.44	4,374.9
PRODHOT	129.10	0.20	0.000	132.29	12,307.2
PRODWAR	129.13	1.00	0.000	132.29	12,307.2
PRODUCT	40.00	1.00	0.000	132.29	12,307.2
NBBOT	153.15	0.30	0.000	2.00	221.3

REACT rejects 15.354 MW isothermally at 300 °C and COOL-1 rejects 4.380 MW between 287 and 120 °C. Both are in the steam-raising range; the quantified case for COOL-1 is in Section 6.3. Report 1 identified recovery of the exotherm as steam as one of the reasons for selecting the vapour-phase route, and discarding it to cooling water would negate that argument.

DEC is excluded because the cooling water cannot reach it. The block operates isothermally at 40 °C while the water returns at 42 °C, and Aspen Plus rejected the assignment with

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TEMPERATURES CROSS WITH UTILITY CW IN DECANter BLOCK DEC
BLOCK TEMPERATURES 3.131497D+02 K AND 3.131500D+02 K
UTILITY TEMPERATURES 3.181500D+02 K AND 3.031500D+02 K

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This was a correct diagnosis of a real design problem and not a modelling error. The duty is 0.176 MW, 0.8 per cent of the load, and shall be handled by a trim exchanger on chilled water or an air-cooled unit.

The same consideration applies with greater force to COOL-2, which also operates at 40 °C but carries 5.572 MW. Supply and return were first set at 30 and 45 °C, a 15 K range, which places the return above the process temperature and is infeasible. They were revised to 30 and 42 °C, a 12 K range, which

Table 12: PRODUCT, 132.29 kmol/h and 12,307.2 kg/h at 40 °C and 1.0 bar

Component	kmol/h	mol %	kg/h	wt %	Report 2, wt %
Aniline	131.351	99.29	12,232.5	99.39	97.04
Cyclohexylamine	0.523	0.40	51.8	0.42	1.06
Nitrobenzene	0.147	0.11	18.0	0.15	0.07
Water	0.269	0.20	4.8	0.04	1.83
Total	132.29	100.00	12,307.2	100.00	

Table 13: Block duties and assigned utilities

Block	Service	cal/s	MW	Utility
VAP-1	Feed vaporiser	+874,407	+3.659	HPSTEAM
FEHE	Feed-effluent exchanger	50,324	0.211	Internal
REACT	Reactor, isothermal	−3,669,783	−15.354	Unassigned, steam raising
COOL-1	Effluent cooler to 120 °C	−1,046,828	−4.380	Unassigned, steam raising
COOL-2	Effluent cooler to 40 °C	−1,331,805	−5.572	CW
FLASH	Flash drum	+5,188	+0.022	Negligible
DEC	Decanter	−42,163	−0.176	Chilled water or air cooler
ARCOND	Stripper overhead condenser	−1,830,523	−7.659	CW
COL-1	Condenser	−660,333	−2.763	CW
COL-1	Reboiler	+931,068	+3.896	MPSTEAM
COL-2	Condenser	−1,271,875	−5.322	CW
COL-2	Reboiler	+1,159,755	+4.852	LPSTEAM
STRIP	Reboiler	+1,905,416	+7.972	LPSTEAM
PRODCOOL	Product cooler	−158,760	−0.664	CW
COMP	Shaft work	—	+0.278	Electrical
P-1, P-2	Shaft work	—	+0.0004	Electrical

retains COOL-2 on the main header at a 2 K approach. The narrower range raises the circulation, and the trade is discussed in Section 4.4.

4.2 Circulation rate and verification

With water supplied at 30 °C and returned at 42 °C and a mean specific heat of 4.178 kJ/kg·K,

$$\dot{m} = \frac{Q}{c_p \Delta T} = \frac{21.980 \times 10^6}{4178 \times 12} = 438.4 \text{ kg/s} = 1578 \text{ t/h}$$

The loop was designed by hand and then computed independently in Aspen Plus by defining cooling water as a utility and assigning it to each block, so that the two may be compared.

The two agree to 0.39 per cent overall and better than 0.5 per cent on every service. The residual arises from the mean specific heat, Aspen Plus evaluating it from the water property model across the actual range rather than at a single mean temperature.

4.3 Tower water balance

The tower balance is established by hand, no facility for it existing in Aspen Plus. With a mean latent heat of 2418 kJ/kg, four cycles of concentration and drift at 0.02 per cent of circulation:

Table 14: Cooling water services, hand calculation against Aspen Plus

Service	Hot side	Duty, MW	Hand, t/h	Aspen, t/h	Deviation, %
ARCOND	97.5 to 40 °C	7.659	550.0	552.1	0.38
COOL-2	120 to 40 °C	5.572	400.1	401.7	0.40
COL-2 condenser	129.1 °C condensing	5.322	382.2	383.6	0.37
COL-1 condenser	66.9 °C condensing	2.763	198.4	199.2	0.40
PRODCOOL	129 to 40 °C	0.664	47.7	47.9	0.42
Total		21.980	1578.3	1584.5	0.39

$$E = \frac{Q}{\lambda} = \frac{21.980 \times 10^6}{2.418 \times 10^6} = 9.09 \text{ kg/s} = 32.7 \text{ t/h} \quad (1)$$

$$D = 0.0002 \times 1578 = 0.32 \text{ t/h} \quad (2)$$

$$B = \frac{E}{N-1} - D = \frac{32.7}{3} - 0.32 = 10.6 \text{ t/h} \quad (3)$$

$$M = E + D + B = \frac{NE}{N-1} = 43.6 \text{ t/h} \quad (4)$$

Equation (3) follows from the solids balance. If N is the ratio of dissolved solids in the circulating water to that in the make-up, then $Mx_M = (B + D)Nx_M$; eliminating M between this and the water balance $M = E + D + B$ gives the result.

Two checks are available. The evaporation of 32.7 t/h is 2.07 per cent of circulation, against the customary rule of about one per cent per 5.6 K of range, which for 12 K gives 2.14 per cent, agreeing within 3 per cent. Separately the closed form $M = NE/(N-1) = 4 \times 32.7/3 = 43.6 \text{ t/h}$ reproduces the make-up obtained by summing the three losses.

The figure is conservative in one respect: not all the tower duty leaves as latent heat, typically 10 to 25 per cent being transferred to the air as sensible heat and more in winter, so the evaporation and make-up are upper bounds. Make-up at 43.6 t/h is 2.8 per cent of circulation and 349,000 t/y over 8000 hours.

4.4 Equipment and design conditions

Pump power, with $Q = \dot{m}/\rho = 438.4/993 = 0.4415 \text{ m}^3/\text{s}$, $H = 35 \text{ m}$ and $\rho = 993 \text{ kg/m}^3$:

$$P_{\text{hyd}} = \rho g H Q = 993 \times 9.81 \times 35 \times 0.4415 = 150.5 \text{ kW}, \quad P_{\text{elec}} = \frac{150.5}{0.75 \times 0.95} = 211 \text{ kW}$$

Fan power, at $L/G = 1.2$ giving an air rate of $438.4/1.2 = 365.3 \text{ kg/s}$, that is $317.7 \text{ m}^3/\text{s}$ at 1.15 kg/m^3 , against 180 Pa static:

$$P_{\text{elec}} = \frac{317.7 \times 180}{0.60 \times 0.95} = 100.3 \text{ kW}$$

The 35 m head comprises approximately 10 m of tower lift, 15 m across the exchanger network and 10 m of piping and valve losses, and shall be firmed up once the piping is routed.

Two limitations are recorded. A supply temperature of 30 °C implies a design wet-bulb of about 25 °C at a 5 K approach, whereas the design wet-bulb at Kanpur is 27 to 29 °C in the pre-monsoon period, so 33 to 34 °C is the honest figure for detailed design; the consequence would be to make COOL-2 and FLASH at 40 °C infeasible on cooling water and to move a further 5.6 MW to a chilled or air-cooled duty. Secondly the 12 K range adopted to keep COOL-2 feasible is narrow; a 15 K range would reduce circulation to 1263 t/h and pump power to 169 kW and would be preferred if the 40 °C services were

Table 15: Cooling water loop equipment

Item	Basis	Result
Circulation	21.980 MW, 12 K range	1578 t/h, 1589 m ³ /h
Circulation pump	35 m head, $\eta = 0.75$, motor 0.95	211 kW electrical
Induced-draft fans	$L/G = 1.2$, 180 Pa, $\eta = 0.60$	100 kW electrical
Tower cells	400 m ³ /h per cell	4 duty, 1 standby
Basin	10 minutes holdup	265 m ³
Evaporation / make-up	4 cycles of concentration	32.7 / 43.6 t/h
Total parasitic power		311 kW

moved off the main header in any event. The two questions are linked and shall be resolved together in Report 5.

5 Gas Turbine Captive Power Plant

5.1 Basis and sizing

The purge is the only combustible waste stream in the plant. Its lower heating value is 0.610 MW on the Report 2 flow, of which hydrogen contributes 82 per cent, so it cannot support a power plant alone and a natural gas make-up is provided.

The plant is sized so that the net electrical output matches the plant electrical demand established in Section 6.4. A simple open Brayton cycle with exhaust heat recovery has been adopted, this being appropriate for a unit of this size on a chemical plant.

Air is supplied at 290 kmol/h, corresponding to 200 per cent excess over stoichiometric, selected to hold the turbine inlet temperature near 1150 °C, the limit consistent with uncooled blading in a machine of this class. The value obtained is 1153.85 °C.

An initial case was built at 1207 kmol/h of air and 39.9 kmol/h of natural gas, which gave 2.966 MW net electrical, four times the plant demand. Since the fuel-to-air ratio fixes the flame temperature, and the cycle efficiency is set by pressure ratio and turbine inlet temperature rather than by size, the case was rescaled at constant ratio to 290 and 7.5 kmol/h. The flame temperature moved by 0.5 K only, which confirms the scaling, and no other specification required adjustment.

5.2 Configuration and results

Table 16: Gas turbine blocks and results

Block	Model	Specification	Result
AIRCOMP	Compr	Isentropic, 1 to 12 bar, $\eta_{is} = 0.85$, $\eta_m = 0.98$	+864.9 kW, 377.3 °C
FUELCOMP	Compr	Isentropic, 2.5 to 12 bar, $\eta_{is} = 0.85$	+13.4 kW, 209.5 °C
COMBUST	RGibbs	12 bar, adiabatic, full equilibrium	1153.85 °C
TURBINE	Compr, turbine	Isentropic, 12 to 1 bar, $\eta_{is} = 0.88$	−1617.0 kW, 593.2 °C
HRSG	Heater	150 °C, 1 bar	−1.179 MW

The combustor is an RGibbs block minimising Gibbs free energy at constant pressure with zero heat duty, so the adiabatic flame temperature is an output and not an assumption. The stack temperature of 150 °C is kept well above the water dew point of the flue gas, so as to leave a margin against condensation and corrosion in the stack. No sulphur species are present in this fuel, so an acid dew point does not arise here.

5.3 Fuel and performance

Table 17: Fuel input

Fuel	kmol/h	LHV, MJ/kmol	Thermal, MW	Share, %
Purge, hydrogen	7.435	241.8	0.499	21.9
Purge, benzene	0.119	3135.6	0.104	4.5
Purge, aniline	0.006	3290	0.005	0.2
Purge, ammonia	0.017	316.8	0.001	0.1
Purge subtotal	8.31		0.610	26.7
Natural gas, as methane	7.5	802.6	1.672	73.3
Total			2.282	100.0

$$\text{Net shaft power} = 1617.0 - 864.9 - 13.4 = 738.7 \text{ kW} \quad (5)$$

$$\text{Net electrical power} = 738.7 \times 0.96 = 709.1 \text{ kW} \quad (6)$$

$$\eta_{\text{thermal}} = 709.1/2282 = 31.1 \% \quad (7)$$

$$\eta_{\text{CHP}} = (0.709 + 1.179)/2.282 = 82.7 \% \quad (8)$$

The net electrical output of 709.1 kW matches the plant demand of 708.5 kW to 0.08 per cent, which is the design requirement. The simple-cycle efficiency of 31.1 per cent is consistent with commercial machines of this class, which achieve 30 to 33 per cent, so the model has not returned an optimistic figure.

The purge now contributes 26.7 per cent of the fuel input rather than the 6.4 per cent it would have contributed at the original size. The plant is a cogeneration unit in which the waste hydrogen is a material fuel, which is a better description of a captive plant than the oversized case would have allowed.

The HRSG recovers 1.179 MW, raising approximately 1.8 t/h of 10 bar steam from feedwater at 105 °C. Against a plant steam demand of 39.0 t/h this is 4.6 per cent, so the recovered heat is a secondary credit and not a solution to the steam deficit identified in Section 3.8.

Sizing the plant to match demand exactly leaves no margin for start-up, for the miscellaneous allowance in Section 6.4 proving low, or for a fouled tower drawing more fan power. Normal practice would size 10 to 15 per cent above demand and retain a grid connection for standby. The matched case is reported here because it is the design requirement, and the margin is noted as a matter for detailed design.

5.4 Combustion products and balance checks

The flue gas leaves at 302.16 kmol/h and 8529.6 kg/h, comprising 76.0 mol per cent nitrogen, 13.6 per cent oxygen, 7.7 per cent water and 2.7 per cent carbon dioxide, which is a normal lean gas-turbine exhaust. Three checks were made.

Carbon. Methane, benzene and aniline supply $7.5 + 0.119 \times 6 + 0.006 \times 6 = 8.250$ kmol/h of carbon, and the flue gas carries 8.25 kmol/h of carbon dioxide.

Oxygen. $290 \times 0.21 = 60.9$ kmol/h is supplied and 41.23 kmol/h remains, so 19.67 kmol/h is consumed. The stoichiometric requirement, taking 2 mol of oxygen per mol of methane, 0.5 per mol of hydrogen, 7.5 per mol of benzene, 7.75 per mol of aniline and 0.75 per mol of ammonia, is

$$7.5 \times 2 + \frac{7.435}{2} + 0.119 \times 7.5 + 0.006 \times 7.75 + 0.017 \times 0.75 = 19.669 \text{ kmol/h}$$

The residual oxygen corresponds to 200 per cent excess air as specified.

Completeness. Residual methane is of order 10^{-29} kmol/h, hydrogen 10^{-5} and ammonia 10^{-12} , confirming that the excess air is adequate and no unburnt fuel leaves the combustor.

The carbon dioxide emission is 363 kg/h, that is 2900 t/y, or 0.030 t per tonne of aniline.

Two limitations of the combustion model are recorded. Carbon monoxide was not included in the component slate, so although the equilibrium carbon monoxide at 200 per cent excess air and 1154 °C would be negligible, the model cannot demonstrate it. Nitrogen oxides were likewise not included; the ammonia is fully consumed, but ammonia is fuel-bound nitrogen and converts preferentially to NO_x, which is not controlled by the same measures as thermal NO_x. The flow is small but the point is flagged for the safety and environmental treatment of a later report.

FUELCOMP reports FEED STREAM IS BELOW DEW POINT, the purge at 40 °C and 2.5 bar sitting marginally into the two-phase region. The condensate is a trace but a suction knockout drum belongs on the fuel compressor in the physical plant.

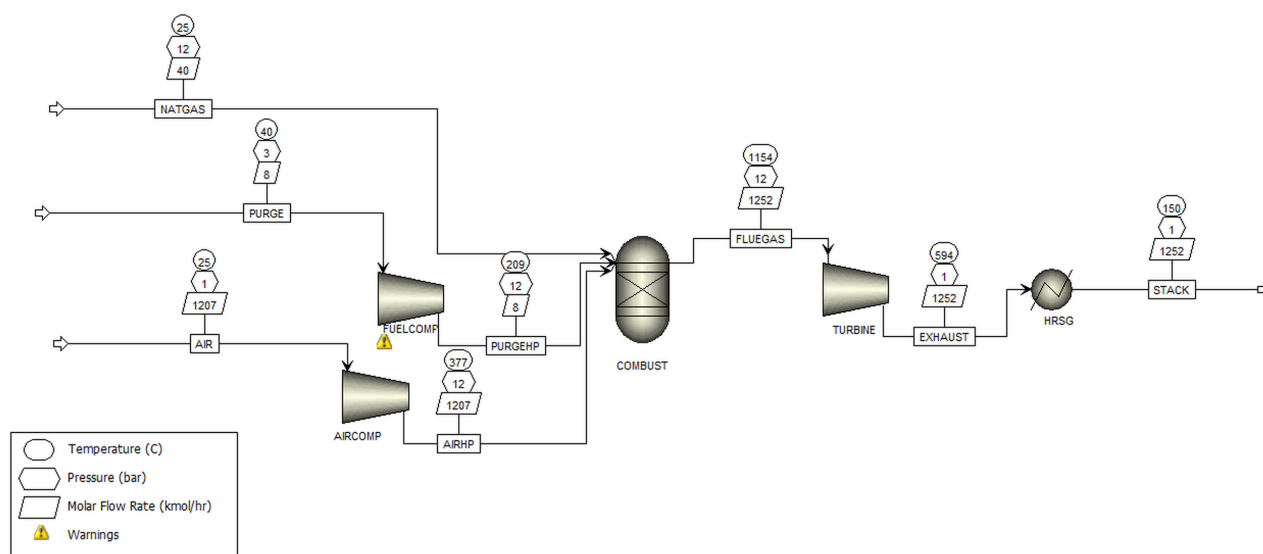


Figure 2: Gas turbine captive power plant. The flows marked on this diagram are those of the first, oversized case at 1207 kmol/h of air and 39.9 kmol/h of natural gas. The design reported in Section 5 is the rescaled one at 290 kmol/h of air and 7.5 kmol/h of natural gas. The block arrangement, pressures and efficiencies are the same in both.

6 Utility Consumption and Study-Level Costing

6.1 Utility definitions

Four utilities were defined in Aspen Plus and assigned to the blocks, so that consumption and cost are computed by the model rather than by hand. Prices are typical Indian industrial figures for a study estimate and are consistent with the basis of Report 1 Section 5.3.

Table 18: Utility definitions

Utility	Type	Conditions	Price	Serves
CW	Water	30 to 42 °C, 4 to 3 bar	0.02 USD/t	Five cooling services
LPSTEAM	Steam	10 bar saturated, 184 °C	0.026 USD/kg	COL-2, STRIP reboilers
MPSTEAM	Steam	16 bar saturated, 201 °C	0.030 USD/kg	COL-1 reboiler
HPSTEAM	Steam	70 bar saturated, 285 °C	0.035 USD/kg	VAP-1

The steam utilities are specified between saturated vapour and saturated liquid at constant pressure, the duty in each case being latent. The level for each service is set by the temperature it must drive:

VAP-1 vaporises at 263 °C and requires 70 bar, COL-1 reboils at 185.95 °C and cannot be served by 10 bar steam at 184 °C, and COL-2 and STRIP at 153.15 and 101.15 °C are comfortably served by 10 bar.

6.2 Consumption and cost

Table 19: Utility consumption and cost from Aspen Plus

Utility	Block	Service	Usage, kg/h	Cost, USD/h
HPSTEAM	VAP-1	Feed vaporiser	8,753	306.35
MPSTEAM	COL-1	Reboiler	7,261	217.82
LPSTEAM	STRIP	Reboiler	14,265	370.90
LPSTEAM	COL-2	Reboiler	8,683	225.75
Steam subtotal			38,962	1120.83
CW	ARCOND	Condenser	552,133	11.04
CW	COOL-2	Effluent cooler	401,711	8.03
CW	COL-2	Condenser	383,631	7.67
CW	COL-1	Condenser	199,179	3.98
CW	PRODCOOL	Product cooler	47,886	0.96
Cooling water subtotal			1,584,540	31.69
Total, thermal utilities				1152.52

Over 8000 hours the thermal utility cost is USD 9.22 million per annum. Adding the electrical demand of Section 6.4 at 0.09 USD/kWh gives **USD 9.73 million per annum**, that is **USD 99.4 per tonne of contained aniline**.

Steam accounts for 97.3 per cent of the thermal utility cost and cooling water for 2.7 per cent. The economic burden lies wholly on the hot side. Two consequences follow: the selection of the lower stripper boilup in Section 2.5 is worth USD 1.06 million per annum on its own and is the most valuable single decision in this report, and effort spent minimising the cooling water circulation has little return, so that loop should be designed for approach temperature and reliability rather than for minimum flow.

6.3 Recommendation on COOL-1

Low-pressure steam delivers heat at $3600/2015 \times 0.026 \times 1000 = 46.5$ USD per MWh. COOL-1 rejects 4.380 MW between 287 and 120 °C, which is in the range where 10 bar steam at 184 °C can be raised. Taking the whole duty as recoverable, a waste-heat boiler in place of a cooling-water exchanger would displace

$$4.380 \times 46.5 = 204 \text{ USD/h} = 1.63 \text{ million USD/y}$$

that is about 18 per cent of the thermal utility bill, and would also reduce the cooling water circulation by 315 t/h and the make-up by 8.7 t/h.

This is a screening figure only. Since the stream is cooled to 120 °C while the steam is raised at 184 °C, the lower part of the duty cannot be recovered at that steam level, and the recoverable fraction has to be fixed by a temperature-profile calculation on the exchanger. That is taken up with the pinch analysis of Report 5. On the strength of the screening figure COOL-1 has been left off the cooling water header. The alternative case, with COOL-1 on cooling water, needs 1893 t/h of circulation and 52.3 t/h of make-up and has been costed for comparison only.

The larger prize is REACT at 15.354 MW, but recovering the exotherm requires the reactor configuration to be settled first and belongs to Report 5.

6.4 Electrical demand

Table 20: Plant electrical demand

Consumer	kW	Source
COMP, recycle gas compressor	278.4	Aspen Plus block result
Cooling water circulation pump	211.3	Section 4.4
Cooling tower fans	100.3	Section 4.4
P-1 and P-2	0.4	Aspen Plus block results
Lighting, instrumentation and miscellaneous, 20 %	118.1	Allowance
Total	708.5	

The captive plant of Section 5 generates 709.1 kW, matching this demand. The electrical cost of USD 0.51 million per annum included in Section 6.2 is therefore avoidable in the integrated case, and Report 1 was correct to treat electricity as embedded rather than purchased.

7 Objectives Which Could Not Be Accomplished, With Reasons

- **The ANILINE–NITRO-01 pair has still not been regressed.** Carried forward from Report 1 Section 7 and Report 2 Section 6. The laboratory time in this period went to the column redesign, which required thirteen successive converged cases, and to the second flowsheet for the power plant. The databank values have A_{ij} and A_{ji} both zero with only the B terms present and a stated validity of 83 to 99 °C, whereas the COL-2 reboiler now runs at 153.15 °C, which is further outside the range than the 144.15 °C of Report 2. The COL-2 results, and the nitrobenzene content of the product in particular, are therefore provisional. This is now the most pressing outstanding item and shall be completed before any further refinement of COL-2.
- **The third azeotrope reported in Report 1 Section 6.8 has not been identified.** Report 1 undertook to pin it down before the sequencing of the present report; an Azeotrope Search over the ternary was not run. The sequencing of Section 2.1 rests on the binary aniline–water azeotrope and the boiling point order, and would be affected only if the third azeotrope lay within the composition range the columns actually traverse. Since COL-1 operates between a water-rich overhead and an essentially binary aniline–nitrobenzene bottoms this is unlikely, but it has not been demonstrated. It shall be run with the regression above.
- **Aniline purity.** 99.39 wt % against 97.04 in Report 2, against a specification of 99.9 wt %. The binding impurity is cyclohexylamine at 0.42 wt %, which cannot be removed in either existing column for the reason established in Section 2.4.
- **Nitrobenzene in the product.** 1466 ppm against a specification of 5 ppm. A sharper cut needs more stages in COL-2 or a third column, and in either case the binary parameters must be regressed first, the present result resting on an extrapolated correlation.
- **Aniline in the wastewater.** 154 ppm against 2135 ppm in Report 2 and a specification of 1 ppm. Section 2.5 establishes that equilibrium stripping cannot reach 1 ppm at any practicable boilup and that a polishing step is required.
- **The aniline leaving in WATOVHD.** WATOVHD carries 2.515 kmol/h of aniline, 234 kg/h, which is 1.9 per cent of production and the largest single loss in the plant. Its composition, $x_{\text{aniline}} = 0.0464$, lies inside the miscibility gap of 0.037 to 0.575, so it is a two-phase stream and is not treatable as wastewater as it stands. Routing it to the stripper or to a second decanter would recover most of the aniline. Neither Report 2 nor the present report examined this and it is recommended for Report 4.

- **The cyclohexylamine in the wastewater.** WASTEWTR carries 1.637 kmol/h of cyclohexylamine, 3.7 wt per cent, which is a larger discharge problem than the residual aniline and was not visible in Report 2 because the stripper load was much smaller. It shall be considered with the wastewater polishing step.
- **Heat integration beyond FEHE.** A pinch analysis of the complete network, including steam raising from REACT and COOL-1 and preheating of the combined rather than the organic reactor feed, is deferred to Report 5 as stated in Report 2.

8 Any Other Challenges

- **The partial condenser had previously crashed the Aspen Plus engine.** The cause was a redundant specification and not a solver weakness. A Partial-Vapor-Liquid condenser produces two distillate streams natively and needs only the ratio between them; a liquid side draw declared in addition over-specifies the column. The engine reported first PRODUCTS NOT COMPLETE, SID = WATOVHD, EITHER FLOW OR SIDEDRAW TO FEED RATIO MUST BE SPECIFIED and then, on a flow being supplied, SEVERE ERROR: SUM OF PRODUCT FLOW SPECIFICATIONS EXCEEDS FEED FLOW, before terminating. Deleting the side draw resolved it.
- **Design specifications have been preferred to sensitivity analyses throughout.** A sensitivity on the stripper boilup ratio attempted in the previous period failed first with ERROR WHILE CHECKING INPUT SPECIFICATIONS FOR VARIED VARIABLE, the block being specified on boilup ratio rather than the varied quantity, and then, on the variable being corrected, with a hundred consecutive FORTRAN DIVIDE BY ZERO ENCOUNTERED errors which terminated execution. The stripper sits inside the recycle loop, so every sensitivity point re-tears and re-converges all three recycles and a step landing on zero internal flow cascades. The boilup study of Section 2.5 was therefore run as discrete converged cases.
- **The Strongly Non-Ideal Liquid algorithm could not be used.** Aspen Plus locks the convergence algorithm to Standard whenever design specifications are active on a RadFrac block. Standard proved adequate once three-phase behaviour was confined to the condenser stage.
- **Parameters tested and abandoned because another variable overshot.** Recorded in the sections concerned and collected here: the COL-1 distillate vapour fraction of 0.10 on the Report 2 feed, which failed to converge; COL-2 reflux ratio 4, which reduced nitrobenzene but raised the reboiler duty 74 per cent for no purity gain; COL-2 bottoms 3 kmol/h, which reduced nitrobenzene 16 per cent while pushing 0.97 kmol/h more aniline into the recycle; stripper boilup 5, which reduced residual aniline 2.7-fold for 5.09 MW of additional duty and 410 kmol/h of additional recycle; and the two cyclohexylamine side draws, which removed water and aniline instead of the impurity.
- **Purge basis used for the power plant.** The power plant has been built on the purge of 8.31 kmol/h taken from Report 2. The converged flowsheet of the present report gives 9.79 kmol/h and 49.0 kg/h, with 8.833 kmol/h of hydrogen, so the purge fuel value is about 0.72 MW instead of 0.610 MW. At the same turbine inlet temperature the natural gas make-up simply comes down by the same amount, so the net power changes by well under one per cent and the sizing of Section 5 holds. The power plant file shall be run again on the new purge in the next period.
- **VENT requires disposal and has not been costed.** The new vapour distillate from COL-1 is 6.02 kmol/h and 262.6 kg/h at 43 mol per cent benzene, 29 per cent ammonia and 25 per cent water. It cannot be released and requires a flare or thermal oxidiser, which is an addition to the equipment list.
- **The work was carried out as thirteen successive files.** Because every change to the liquid train alters the feed to the next unit and all three recycles must re-converge, each change was made singly and saved as che453_R3_00 through che453_R3_12, so that a failure could be rolled back without loss. Two changes required full reinitialisation, the ANRECOV rerouting and the addition of P-2, both altering the topology.

9 Number of Hours Spent During This Period

Approximately **6 hours per member**, that is about **36 person-hours**, against 18 for Report 1 and 24 for Report 2. This covers the redesign of the three columns across thirteen successive converged cases, the construction and convergence of the separate power plant flowsheet, the cooling water design and its verification against the Aspen Plus utility calculation, the utility definitions and costing, and the preparation of this report.

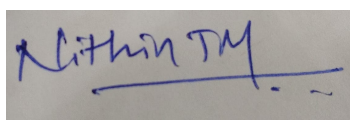
10 Contributions From Individual Members

- **Nithin T M and Aditya Nitin Patil — process flowsheet.** The whole of the work on `che453.R3.12.apw`: the partial condenser configuration and the selection of the distillate vapour fraction, the rerouting of the stripper overhead with the ARCOND condenser, the stage count and feed location of COL-1 and its drying design specification, the diagnosis of the failed aniline recovery specification against the azeotrope, the three-phase condenser, the re-specification of COL-2 on bottoms rate together with the two rejected trials, the stripper boilup study, the two cyclohexylamine side draw trials, the product conditioning and recycle pumps, the convergence work across the thirteen successive cases, and the extraction of the material and energy balances of Section 3.
- **Shubham Singh and Shivam Dongare — captive power plant.** The whole of the second flowsheet: the property method selection, the block configuration, the air rate required to hold the turbine inlet temperature, the initial case and its rescaling to match the plant electrical demand, the fuel split and performance calculations, and the combustion balance checks of Section 5.4.
- **Nithin D H — utilities and costing.** The four utility definitions in Aspen Plus and their assignment to the blocks, the extraction of the consumption and cost results, and the waste heat boiler evaluation for COOL-1 in Section 6.3.
- **Sajag Masane — reference list and proof-reading.**

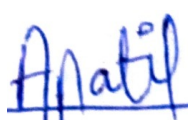
All six members have reviewed the complete report.

11 References

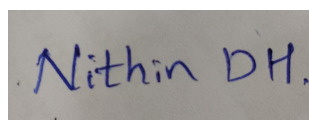
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13. Group 14, CHE453 Bi-weekly Report 1, 9 August 2026.
14. Group 14, CHE453 Bi-weekly Report 2, 23 August 2026.
15. Aspen Plus simulation files `che453.R3.12.apw` and the separate captive power plant file, with the model summary, stream summary and utility results exported as workbooks, 5 September 2026.

Signatures of membersHandwritten signature of Nithin T M in blue ink.

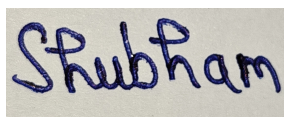
Nithin T M

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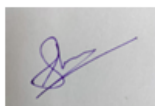
Aditya Nitin Patil

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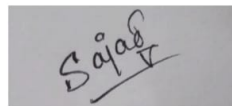
Nithin D H

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Shubham Singh

Handwritten signature of Shivam Dongare in blue ink.

Shivam Dongare

Handwritten signature of Sajag Masane in blue ink.

Sajag Masane